

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. SYSTEM COMMUNICATION PROTOCOLS In the ComicCraft architecture, "Communication" refers to the seamless, asynchronous data exchange between the local FastAPI web server and the external cloud AI services. Since the application relies heavily on third-party generative models, establishing secure and rapid communication channels was a critical development priority.

3. INTER-SYSTEM COMMUNICATION (API CALLS)

- **Text Generation (Google Gemini):** The backend communicates with Google's Generative AI endpoints using RESTful API calls. The system sends HTTP POST requests containing user prompts formatted as strings, and the Gemini API responds with structured JSON data (for the comic outline) and plain text (for the dialogues).
- **Image Generation (Hugging Face):** Communication with the Stable Diffusion model is handled via `urllib.request`. The backend sends a JSON payload containing the scene description and an authorization bearer token. The API responds with raw byte data, which the system decodes and saves as local `.jpg` files.

4. SYSTEM-TO-USER COMMUNICATION (UI FEEDBACK) Effective communication with the end-user is managed through the Jinja2 templating engine.

- **Input State:** The HTML forms clearly communicate the required parameters (Lead Character, Tone, Style).
- **Loading State:** Due to API latency, the system communicates processing times to the user to prevent them from prematurely refreshing the page.
- **Error State:** If an external API fails (e.g., HTTP 503 Service Unavailable), the backend intercepts the error and communicates a fallback visual (a placeholder image) rather than crashing the application entirely.

